

Profile

M.S. candidate in Chemical Engineering at NTUST, specializing in **electrochemistry**, **ultra-low-temperature electrolytes** and **lithium battery reliability / aging**; co-author on a **Nature Materials** submission.

Research & Industry

Long-Life Li Battery Evaluation for Smart Water Meters

Partner: a smart-water-meter battery maker **Project lead**

Screening methodology Designed full-stage test protocols, experimental workflows and statistical validation models.	Advanced analysis EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.
Database & SOP Established the test database and standardized high-throughput screening procedures (SOP).	Training & tech transfer Trained QA and R&D teams to independently run large-batch screening and data interpretation.

Nature Materials Journal submission — battery thermal stability study Status **Under review**

Co-author Responsible for TGA thermal measurement
Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions.

Experience

2025 – Present	Sustainable Electrochemical Energy Development (SEED) Center, NTUST Lab Equipment Procurement & Asset Maintenance Manager Procurement (>NTD 2M): Led procurement of glovebox O₂ sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes, low-temperature baths and TGA platinum pans; also replace instrument consumables. Lab safety: monitor glovebox O₂ / moisture levels and handle sensor faults; manage and inspect gas cylinders, run lab safety walk-throughs and brief newcomers on safety; helped with on-site response to incidents such as a cut from broken glassware and a glovebox catalyst-canister leak. Hazardous materials: handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li-SOCl₂ cells. - Assembled cells and prepared electrodes in a battery dry room, working in cleanroom garments under clean-area gowning and entry procedures. TGA testing service: Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST — over 600 samples in total, including my own lab's. Instrument maintenance: Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500.
Summer 2022	New Taipei City Dept. of Urban Development Summer Intern, Housing Development Section Reviewed rental subsidy eligibility, cross-checking property, income, and insurance records across agencies (tax bureau, land registry, NHI); handled frontline public inquiries and disputes.
2021 – 2024	Tunghai University Student Housing / GEPT Resident Assistant (2 yrs) / Exam Staff (part-time) Oversaw safety and order in a 100-resident dormitory, mediating disputes and tracking facility safety issues with the dorm supervisor; as part-time exam staff, handled test-paper check-in with zero errors.

Conferences & Training

Presenter	2026 TwiChE Annual Meeting 2026/12/05–06 • Feng Chia University Title: <i>Weakly Solvating Allyl Ether Electrolytes for Subzero Li SPAN Batteries: Solvation Structure, Transport Properties, and Separator Wettability</i>
Vendor seminar	TA Instruments ARES-G3 launch & applications forum 2026/07/17 • National Taiwan University
Vendor lecture	TA Instruments TGA lecture 2026/06/03 • National Taiwan University

Skills & Instruments

Electrochemistry & batteries - Impedance spectroscopy (EIS) - Battery aging analysis - Reliability analysis - Low-temp electrolytes
Characterization - Thermal analysis (TGA / DSC) - Synchrotron (TLS 20A / TPS 27A / TPS 45A) - Rheology (ARES-G3) - Viscometer (DMA 5002/Lovis 2001)
Process & operations - Lithography (nanoimprint / self-assembly) - Equipment procurement & maintenance

Software - Aspen Plus V14 - Aspen Dynamics - Microsoft Office - OriginPro - TA Universal Analysis - TA TRIOS

Education

2025 – 2027 (expected)	 National Taiwan University of Science and Technology M.S. in Chemical Engineering Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.
2020 – 2024	 Tunghai University B.S. in Chemical & Materials Engineering Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).

EHS Licenses

	Dedicated Wastewater and Sewage Treatment Specialist (Class B) 2026 / 06 • certified • No. GB040231
	Indoor Air Quality Maintenance Management Specialist 2026 / 08 • certified • No. IA020236
	Waste Disposal Technician (Class A) 2026 / 10 • certified • No. HA160629
	Occupational Safety and Health Management Personnel (Level B) 2026 / 11 • mid-Nov exam scheduled
	Occupational Safety Management Specialist (Level A) 2027 / 03 • exam scheduled

Instruments

Maintenance - TGA Q500 - DSC 2500
Operation - Anton Paar DMA 5002/Lovis 2001 - ARES-G3 - Glovebox
Procurement - Glovebox O₂ sensor - DSC sample press - Hydraulic electrode punch - Vacuum pump - Eppendorf Pipette - Low-temperature bath